## ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** $144^\circ$ | One part $= 360^\circ \div 15 = 24^\circ$. |
| 2 | **(d)** $7$ | Opposite angles are equal. |
| 3 | **(c)** $20^\circ$ | Adjacent angles add up to $180^\circ$. |
| 4 | **(a)** $28$ cm | Diagonals bisect each other, so $AC = 2 \times OA$. |
| 5 | **(d)** parallelogram | A test for a parallelogram. |
| 6 | **(c)** $25^\circ$ | A diagonal of a rhombus bisects the angle at the vertex. |
| 7 | **(a)** $43$ cm | Each new side is half of a side of the original triangle. |
| 8 | **(c)** $32$ cm$^2$ | Area $= \frac{1}{2}d^2$. |
| 9 | **(d)** $180^\circ$ | One part $= 360^\circ \div 12 = 30^\circ$. |
| 10 | **(b)** $29$ | Opposite angles are equal. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\angle C = \angle A = 88^\circ$ (opposite angles); $\angle B = 180^\circ - 88^\circ = 92^\circ$ (adjacent angles) (1); $\angle D = \angle B = 92^\circ$ (1). | 1+1 |
| 12 | Diagonal $= \sqrt{8^2 + 15^2}$; $= \sqrt{289} = 17$ cm (1); Perimeter $= 2(8 + 15) = 46$ cm (1). | 1+1 |
| 13 | $EF = 17$ cm, $FD = 25$ cm, $DE = 12$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 17 + 25 + 12 = 54$ cm (1). | 1+1 |
| 14 | $\angle C = \angle A = 40^\circ$ (opposite angles); $\angle B = 180^\circ - 40^\circ = 140^\circ$ (adjacent angles) (1); $\angle D = \angle B = 140^\circ$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Opposite angles are equal; $2x + 8 = x + 16$ gives $x = 8$ (1); Angles $A = C = 24^\circ$ (1); $B = D = 180^\circ - 24^\circ = 156^\circ$ (1). | 1+1+1 |
| 16 | Adjacent angles add up to $180^\circ$: $x + 8x = 180^\circ$ (1); $x = 20^\circ$ and the other is $160^\circ$ (1); Opposite angles are equal: the four angles are $20^\circ, 160^\circ, 20^\circ, 160^\circ$ (1). | 1+1+1 |
| 17 | Diagonals bisect each other, so $OA = OC$; $4x + 2 = 3x + 4$ gives $x = 2$ (1); $OA = 10$ cm (1); $AC = 2 \times 10 = 20$ cm (1). | 1+1+1 |
| | **OR:** Opposite angles are equal; $2x + 2 = x + 15$ gives $x = 13$ (1); Angles $A = C = 28^\circ$ (1); $B = D = 180^\circ - 28^\circ = 152^\circ$ (1). | 1+1+1 |
| 18 | Adjacent angles add up to $180^\circ$: $x + 9x = 180^\circ$ (1); $x = 18^\circ$ and the other is $162^\circ$ (1); Opposite angles are equal: the four angles are $18^\circ, 162^\circ, 18^\circ, 162^\circ$ (1). | 1+1+1 |
| | **OR:** Diagonals bisect each other, so $OA = OC$; $5x + 3 = 3x + 13$ gives $x = 5$ (1); $OA = 28$ cm (1); $AC = 2 \times 28 = 56$ cm (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $AD = 14$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 19$ cm (1); Hence $AE = EC$ (2). | 1+1+1+2 |
| | **OR:** In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 8.5$ cm and $QR = PS = 7.5$ cm (1); Perimeter $= 17 + 15 = 32$ cm (2). | 1+1+1+2 |
| 20 | $AD = 13$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 16$ cm (1); Hence $AE = EC$ (2). | 1+1+1+2 |
| | **OR:** In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 6.5$ cm and $QR = PS = 8$ cm (1); Perimeter $= 13 + 16 = 29$ cm (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** $30^\circ$ | Adjacent angles add up to $180^\circ$. |
| 2 | **(d)** 30 cm | Diagonals bisect each other, so $AC = 2 \times OA$. |
| 3 | **(b)** parallel to the third side and half of it | Midpoint theorem. |
| 4 | **(a)** $57^\circ$ | A diagonal of a rhombus bisects the angle at the vertex. |
| 5 | **(b)** 48 cm | Each new side is half of a side of the original triangle. |
| 6 | **(a)** 50 $cm^2$ | Area $= \frac{1}{2}d^2$. |
| 7 | **(c)** $144^\circ$ | One part $= 360^\circ \div 15 = 24^\circ$. |
| 8 | **(a)** 24 | Opposite angles are equal. |
| 9 | **(b)** $45^\circ$ | Adjacent angles add up to $180^\circ$. |
| 10 | **(d)** 28 cm | Diagonals bisect each other, so $AC = 2 \times OA$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Diagonal $= \sqrt{10^2 + 24^2}$; $= \sqrt{676} = 26$ cm (1); Perimeter $= 2(10 + 24) = 68$ cm (1). | 1+1 |
| 12 | $EF = 12$ cm, $FD = 9$ cm, $DE = 17$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 12 + 9 + 17 = 38$ cm (1). | 1+1 |
| 13 | $\angle C = \angle A = 126^\circ$ (opposite angles); $\angle B = 180^\circ - 126^\circ = 54^\circ$ (adjacent angles) (1); $\angle D = \angle B = 54^\circ$ (1). | 1+1 |
| 14 | Diagonal $= \sqrt{16^2 + 30^2}$; $= \sqrt{1156} = 34$ cm (1); Perimeter $= 2(16 + 30) = 92$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Opposite angles are equal; $4x + 10 = x + 61$ gives $x = 17$ (1); Angles $A = C = 78^\circ$ (1); $B = D = 180^\circ - 78^\circ = 102^\circ$ (1). | 1+1+1 |
| 16 | Adjacent angles add up to $180^\circ$: $x + 4x = 180^\circ$ (1); $x = 36^\circ$ and the other is $144^\circ$ (1); Opposite angles are equal: the four angles are $36^\circ, 144^\circ, 36^\circ, 144^\circ$ (1). | 1+1+1 |
| 17 | Diagonals bisect each other, so $OA = OC$; $4x + 4 = 1x + 31$ gives $x = 9$ (1); $OA = 40$ cm (1); $AC = 2 \times 40 = 80$ cm (1). | 1+1+1 |
| | **OR:** Opposite angles are equal; $3x + 4 = x + 30$ gives $x = 13$ (1); Angles $A = C = 43^\circ$ (1); $B = D = 180^\circ - 43^\circ = 137^\circ$ (1). | 1+1+1 |
| 18 | Adjacent angles add up to $180^\circ$: $x + 5x = 180^\circ$ (1); $x = 30^\circ$ and the other is $150^\circ$ (1); Opposite angles are equal: the four angles are $30^\circ, 150^\circ, 30^\circ, 150^\circ$ (1). | 1+1+1 |
| | **OR:** Diagonals bisect each other, so $OA = OC$; $4x + 1 = 2x + 9$ gives $x = 4$ (1); $OA = 17$ cm (1); $AC = 2 \times 17 = 34$ cm (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $AD = 23$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 17$ cm (1); Hence $AE = EC$ (2). | 1+1+1+2 |
| | **OR:** In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 15$ cm and $QR = PS = 7.5$ cm (1); Perimeter $= 30 + 15 = 45$ cm (2). | 1+1+1+2 |
| 20 | $AD = 8$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 13$ cm (1); Hence $AE = EC$ (2). | 1+1+1+2 |
| | **OR:** In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 5.5$ cm and $QR = PS = 9$ cm (1); Perimeter $= 11 + 18 = 29$ cm (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** are equal and bisect each other | Property of a rectangle. |
| 2 | **(b)** $33^\circ$ | A diagonal of a rhombus bisects the angle at the vertex. |
| 3 | **(c)** 29 cm | Each new side is half of a side of the original triangle. |
| 4 | **(d)** 128 $cm^2$ | Area $= \frac{1}{2}d^2$. |
| 5 | **(c)** $144^\circ$ | One part $= 360^\circ \div 15 = 24^\circ$. |
| 6 | **(a)** 22 | Opposite angles are equal. |
| 7 | **(d)** $60^\circ$ | Adjacent angles add up to $180^\circ$. |
| 8 | **(b)** 34 cm | Diagonals bisect each other, so $AC = 2 \times OA$. |
| 9 | **(a)** $360^\circ$ | Split into two triangles. |
| 10 | **(d)** $28^\circ$ | A diagonal of a rhombus bisects the angle at the vertex. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $EF = 13$ cm, $FD = 24$ cm, $DE = 18$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 13 + 24 + 18 = 55$ cm (1). | 1+1 |
| 12 | $\angle C = \angle A = 60^\circ$ (opposite angles); $\angle B = 180^\circ - 60^\circ = 120^\circ$ (adjacent angles) (1); $\angle D = \angle B = 120^\circ$ (1). | 1+1 |
| 13 | Diagonal $= \sqrt{5^2 + 12^2}$; $= \sqrt{169} = 13$ cm (1); Perimeter $= 2(5 + 12) = 34$ cm (1). | 1+1 |
| 14 | $EF = 12$ cm, $FD = 10$ cm, $DE = 21$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 12 + 10 + 21 = 43$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Opposite angles are equal; $2x + 10 = x + 27$ gives $x = 17$ (1); Angles $A = C = 44^\circ$ (1); $B = D = 180^\circ - 44^\circ = 136^\circ$ (1). | 1+1+1 |
| 16 | Adjacent angles add up to $180^\circ$: $x + 3x = 180^\circ$ (1); $x = 45^\circ$ and the other is $135^\circ$ (1); Opposite angles are equal: the four angles are $45^\circ, 135^\circ, 45^\circ, 135^\circ$ (1). | 1+1+1 |
| 17 | Diagonals bisect each other, so $OA = OC$; $4x + 9 = 2x + 23$ gives $x = 7$ (1); $OA = 37$ cm (1); $AC = 2 \times 37 = 74$ cm (1). | 1+1+1 |
|  | **OR:** Opposite angles are equal; $4x + 7 = x + 76$ gives $x = 23$ (1); Angles $A = C = 99^\circ$ (1); $B = D = 180^\circ - 99^\circ = 81^\circ$ (1). | 1+1+1 |
| 18 | Adjacent angles add up to $180^\circ$: $x + 2x = 180^\circ$ (1); $x = 60^\circ$ and the other is $120^\circ$ (1); Opposite angles are equal: the four angles are $60^\circ, 120^\circ, 60^\circ, 120^\circ$ (1). | 1+1+1 |
|  | **OR:** Diagonals bisect each other, so $OA = OC$; $3x + 8 = 4x + 6$ gives $x = 2$ (1); $OA = 14$ cm (1); $AC = 2 \times 14 = 28$ cm (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $AD = 11$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 12$ cm (1); Hence $AE = EC$ (2). | 1+1+1+2 |
|  | **OR:** In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 6$ cm and $QR = PS = 8$ cm (1); Perimeter $= 12 + 16 = 28$ cm (2). | 1+1+1+2 |
| 20 | $AD = 8$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 15$ cm (1); Hence $AE = EC$ (2). | 1+1+1+2 |

| | | |
|---|---|---|
| | **OR:** In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 9$ cm and $QR = PS = 14.5$ cm (1); Perimeter $= 18 + 29 = 47$ cm (2). | 1+1+1+2 |

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